

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2006

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SHOOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?

- A. Plasma
- B. White blood cells
- C. Red blood cells
- D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]
11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2006.

1. Sweat glands in the skin of human are located in the
 - A. Epidermis
 - B. Dermis
 - C. Pores
 - D. Oil glands
2. Which one of the following parts of a human body produces urine?
 - A. Bladder
 - B. Ureter
 - C. Kidney
 - D. Urethra
3. The diagram below represents the reproductive system in the female human body
Fertilization takes place in the body part labeled
 - A. K
 - B. L
 - C. M
 - D. N
4. The following are some of physical changes that take place during adolescence
 - (i) Hips become broader
 - (ii) Chest and shoulders become broader
 - (iii) Pimple may appear on the face
 - (iv) Voice breaks
 - (v) Pubic hair grow

Which of the changes takes place in the girls?

- A. (i) (ii) (v)
 - B. (i) (iii) (iv)
 - C. (ii) (iv) (v)
 - D. (i) (iii) (v)
5. Which one of the following groups consists of only disease that are immunisable?
 - A. Whooping cough, tuberculosis, tetanus
 - B. Measles, malaria, diphtheria
 - C. Dysentery, hepatitis B, poliomyelitis
Bilharzias, yellow fever, cholera

6. The following are stages of HIV infection

- i) Window stage
- ii) Symptoms
 - iii) Full blown
 - iv) Incubation

Which of the following is the correct order of stages?

- A. (ii) (i) (iii) (iv)
- B. (iv) (ii) (iii) (i)
- C. (i) (iv) (i) (iii)
- D. (i) (iii) (iv) (ii)

7. The following are effects of drugs abuse

- i) Rape
- ii) Addiction
- iii) Lack of concentration
- iv) Truancy
- v) Impaired judgment

Which of them are all health effects

- A. (i) (ii) (iii)
- B. (ii) (iii) (iv)
- C. (i) (iv) (v)
- D. (ii) (iii) (v)

8. One evening, pupil noticed a bright heavenly body that was not twinkling. The heavenly body was most likely

- A. Saturn
- B. Uranus
- C. Venus
- D. Neptune

9. The reason for using a biro pen cap as a pivot in the construction of an improvised wind vane is to make the arrow

- A. Rotate freely
- B. Balance on the stand
- C. Tail show the direction of wind
- D. Head heavy

10. Which of the following statements is true about the planet Mars? It is

- A. Nearest to the sun
- B. The brightest planet
- C. Nearest to the earth
- D. The smallest planet

11. Which one of the following is a leguminous plant?
 - A. Cabbage
 - B. Maize
 - C. Groundnuts
 - D. Tomato
12. Fertilization in plants takes place when
 - A. A pollen tubes enters the ovule
 - B. Reproductive cells in pollen grain and ovule fuse
 - C. A pollen grain falls on the stigma and germination
 - D. A pollen tubes enters the ovary
13. which one of the following groups only consists of non- flowering plants
 - A. Pine, cedar, cypress
 - B. Sugarcane, sisal, onion
 - C. Cedar, moss, kale
 - D. Carrot, cassava, carrot
14. Which one of the following groups of plants have the same type of roots system
 - A. Tomato, maize, peas
 - B. Beans, blackjack, cabbage
 - C. Groundnuts, rice, sugarcane
 - D. Millet, cassava, carrot
15. Which one of the following parasites destroys wool in sheep?
 - A. Mites
 - B. Lice
 - C. Fleas
 - D. Ticks
16. The toothless gap in the lower jaw of herbivores is for
 - A. Turning plants materials in the mouth for proper chewing
 - B. Holding food material
 - C. Allowing space for more food
 - D. Preventing the animals from biting their tongues
17. Which one of the following animals has the same number of legs as the tick?
 - A. Weevil
 - B. Crab
 - C. Scorpion
 - D. Cockroach
18. Which one of the following is filter feeder?
 - A. Kingfishers
 - B. Hawk
 - C. Duck
 - D. Sunbird

19. Which one of the following is not an immediate sign of ill health in livestock?
- A. Coughing
 - B. Blood in stool
 - C. Rough coat
 - D. Loss of weight
20. Which one of the following method of softening hard water?
- A. Filtering
 - B. Boiling
 - C. Decanting
 - D. Sieving
21. Which one of the following practices will help to prevent the spread of cholera?
- A. Spraying chemicals in the stagnate water
 - B. Wearing protective shoes when passing through water
 - C. Drinking filtered water
 - D. Covering pit latrines after use
22. Which one of the following NEED NOT be the same when investigating drainage in different type of soil
- A. Amount of the soil in the funnel
 - B. Size of the funnel
 - C. Size of the bottles collecting the draining water
 - D. Amount of water added to the funnel
23. The following are effects of soil erosion
- i) Small holes in the ground
 - ii) Top layer of the soil missing
 - iii) Small and shallow channels in the ground
 - iv) V-shaped trenches in the ground

Which effects are as a results of splash and rill erosion?

- A. (I) and (iv)
 - B. (I) and (iii)
 - C. (ii) and (iii)
 - D. (ii) and (iv)
24. Which one of the following liquids mix?
- A. Water and milk
 - B. Milk and kerosene
 - C. Water and kerosene
 - D. Cooking oil and milk
25. The diagram below represents a set up that can be used to demonstrate a certain property of metals
- Which property of metal was being demonstrated?
- A. Metals expands when heated

- B. Metals bend when heated
 - C. Solid conduct heat
 - D. Hot metals are pulled down by weight
26. A pupil investigate a certain aspect of pressure in used a set up shown in the diagram below
What conclusion could the pupil make from the investigation?
- A. Air exert pressure on the surface of the liquid
 - B. Pressure in the liquid increase with depth
 - C. Pressure in liquid is equal at the same depth
 - D. Pressure in liquid is exerted on the sides only
27. The temperature at which a substance freezes is the same temperature at which the substance
- A. Condense
 - B. Evaporates
 - C. Boils
 - D. Melts
28. Which one of the following consists only of substances that have no definite shape but have definite volume?
- A. Stone, kerosene, carbon dioxide
 - B. Oil, toothpaste, glue
 - C. Clay, flour, cement
 - D. Nitrogen, oxygen water vapour
29. The diagram below represents a set up that can be used to demonstrate heat transfer in air
For the demonstration to work a burning candle must be placed at
- A. P
 - B. Q
 - C. R
 - D. S
30. The percentage of the component of air that is fixed by a bacteria root nodules to make nutrients for plants is
- A. 0.03%
 - B. 0.97%
 - C. 21%
 - D. 78%
31. Which one of the following is the last step when separating mixtures of sand and salt?
- A. Dissolving
 - B. Filtering
 - C. Evaporation
 - D. Decanting
32. The diagram below shows a stone being thrown using a catapult.
The form of energy in a rubber band before the stone is released is
- A. Kinetic

- B. Potential
- C. Chemical
- D. Sound

33. Which one of the following is correct about a solar heater? A solar heater changes
- A. Heat energy to electrical energy
 - B. Light energy to electrical energy
 - C. Chemical energy to heat energy
 - D. Light energy to heat energy
34. Tall building are fitted with sharp pointed metal rods at their tops and connected to the ground. the rods are for
- A. Protecting the building against the lightening
 - B. Protecting electricity users from being electrocuted
 - C. Allowing expansion of walls during hot weather
 - D. Fixing aerals for redios and television
35. Which one of the following is true about the rainbow? It forms
- A. Before heavy rains
 - B. When it rains when the sun is shining
 - C. When the sky is covered with thick clouds
 - D. When it rains in the afternoon
36. Which one of the following energy transformation takes place when the radio that uses cells is operating?
- A. Chemical-> electrical->magnetic->kinetic->sound
 - B. Magnetic->electrical->kinetic->chemical->sound
 - C. Electrical->chemical->kinetic->magnetic->sound
 - D. Kinetic->chemical->electrical->magnetic
37. Which one of the following sources of energy does not contain chemical energy?
- A. Car battery
 - B. Bicycle dynamo
 - C. Food
 - D. Firewood
38. Heat reaches the feet of the person seated by side of the fire by
- A. Conduction and convection
 - B. Radiation only
 - C. Convection only
 - D. Radiation and convection
39. Which one of the following does not pollute air
- A. Exhausted gases from aeroplanes
 - B. Smoke from burning charcoal
 - C. Smoke from the burning domestic waste
 - D. Gases from decaying animals

40. The main reason why it is advisable to spray crops in the direction of the wind is avoid
- A. The chemical getting into contact with the skin
 - B. Wastage of chemical
 - C. Polluting the air
 - D. Inhaling the chemical

41. The following signs and symptoms were observed in a certain child
- i) Thin and weak
 - ii) Wrinkled face
 - iii) Pot bellied
 - iv) Hungry and crying a lot

The child was most likely suffering from

- A. Anemia
 - B. Rickets
 - C. Kwashiorkor
 - D. Marasmus
42. Which one of the following groups consists only of foodstuff that protect the body against disease
- A. Lemon, cabbage, carrot
 - B. Beans, groundnuts, fish
 - C. Maize, potatoes, rice
 - D. Meat, orange, spinach
43. The purpose of salt in the food preservation is to
- A. Remove water
 - B. Add flavor
 - C. Remove air
 - D. Lower temperature
44. The most likely reason why some expectant mothers crave for soil is to
- A. INCREASE breast milk production after delivery
 - B. Increase appetite
 - C. Help in digestion of food
 - D. Provide sufficient mineral nutrients
45. Which one of the following levers when in use has the position of the effort between the load and fulcrum?
- A. Spade
 - B. Wheel barrow
 - C. Claw hammer
 - D. Crowbar
46. A single fixed pulley
- A. Changes direction of force

- B. Increase effort required
- C. Reduce effort required
- D. Reduce load distance

47. The diagram below represents a tyre of a vehicle

The purpose of treads in the vehicle tyre is to

- A. Make them last longer
- B. Increase grip on the road
- C. Reduce surface area
- D. Make them lighter

48. The following are activities carried out when making a beam balance but not in correct order

- i) Suspend the arm to find balance point and mark
- ii) Make a base, stand and the arm
- iii) Fix the stand and the arm to the base
- iv) Suspend tins on both sides of the arm

Which one of the following is the correct order in which the activities are carried out?

- A. (iii) (i) (ii) (iv)
- B. (ii) (i) (iii) (iv)
- C. (iii) (ii) (iv) (i)
- D. (ii) (i) (iv) (iii)

49. Which one of the following statement is correct about a stone thrown a few meters directly upwards by passenger in a moving lorry?

The stone will fall

- A. Infront of the person
- B. Behind the person
- C. On the person
- D. Behind the lorry

50. The diagram below represents a certain weed.

the weed is

- A. Pigweed
- B. Mexican marigold
- C. Black jack
- D. Oxalis